USN | | | | | | | | | | [22MTAIDE51]

THIRD SEMESTER M.Tech EXAMINATION, DECEMBER - 2024

INDUSTRIAL AI

Time:3 Hours **Maximum Marks:100**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries20 marks.
- **ii.** Missing data may be suitably assumed.

1.	**a.**	State the potential barriers to widespread adoption of AI in industrial processes.	CO1	L4	**10**
	b.	Elaborate on case studies of successful Industrial AI applications in sectors such as manufacturing, energy, and transportation.	CO1	L5	**10**
2.	**a.**	Explain the role of historical data repositories in industrial data analytics.	CO2	L2	**10**
	b.	Make use of the common data sources utilized in industrial environments.	CO2	L3	**10**
3.	**a.**	Explain supervised learning techniques used in industrial AI, such as regression, types, and time-series forecasting, and discuss their applications.	CO3	L2	**10**
	b.	Compare different unsupervised learning techniques like clustering, anomaly detection, and dimensionality reduction, and analyze their use cases in industrial contexts.	CO3	L2	**10**
		OR			
4.	**a.**	Identify deep learning architectures, including neural networks, convolutional neural networks (CNNs), recurrent neural networks (RNNs), and transformer models, and explain their relevance in industrial AI applications.	CO3	L2	**20**
5.	**a.**	Compare AI techniques for process optimization, including reinforcement learning, genetic algorithms, and simulated annealing, and analyze their effectiveness in industrial applications.	CO4	L2	**10**
	b.	Differentiate between model-based and model-free approaches to control systems optimization, and discuss scenarios where each approach would be most beneficial.	CO4	L2	**10**
		OR			
6.	**a.**	Explain the role of predictive maintenance and asset management in industrial AI, and evaluate techniques such as prognostics, health monitoring, and failure prediction.	CO4	L2	**10**
	b.	Explain how real-time decision-making can be achieved in industrial environments using AI technologies, and assess the challenges associated with implementing such systems.	CO4	L2	**10**

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7.	**a.**	Evaluate the safety considerations in deploying AI systems in industrial settings, including risk assessment, fail-safe mechanisms, and human-in-the-loop approaches.	CO5	L5	**20**
		OR			
8.	**a.**	Analyze the role of autonomous systems and robotics in AI-driven automation within manufacturing and logistics, and evaluate their impact on operational efficiency.	CO5	L4	**20**